

POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

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Chapter 3: Integral Calculus

CALCULUS 1

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Integral Calculus

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Integral Calculus

1 3.1. The Definite Integral

2 3.2. Improper Integrals

3.1.1. Definition

Definition

Let the function $f(x)$ be defined on $[a, b]$.

+ Divide P the closed interval $[a, b]$ by n points

$$x_0 = a < x_1 < \dots < x_n = b.$$

Set $\Delta_{x_i} = x_i - x_{i-1} \quad \forall i = 1, 2, \dots, n, \Delta_P = \max\{\Delta_{x_1}, \Delta_{x_2}, \dots, \Delta_{x_n}\}$ (it is called *length* of P)

+ Choose an arbitrary point (denotation ξ : goxi) $\xi_i \in [x_{i-1}, x_i]$, $i = 1, 2, \dots, n$. Then,

$$\sigma_n = \sum_{k=1}^n f(\xi_k) \Delta_{x_k} = f(\xi_1) \Delta_{x_1} + f(\xi_2) \Delta_{x_2} + \dots + f(\xi_n) \Delta_{x_n}$$

is called to be *Riemann sum* (after the German mathematician *Bernhard Riemann 1826–1866*) of $f(x)$ on $[a, b]$.

3.1.1. Definition

Definition

Consider the limit

$$I = \lim_{\Delta_P \rightarrow 0} \sigma_n.$$

If there exists the definite limit I , it does not depend P and choosing ξ_i $i = 1, 2, \dots, n$, then I is called *the definite integral* of $f(x)$ on $[a, b]$

and denoted by $\int_a^b f(x)dx$. The symbol $\int$ was introduced by Leibniz and is called an integral sign. It is an elongated S and was chosen because an integral is a limit of sums. Constant a is called the lower limit and b is the upper limit.

Remark

Remark 1

In the special case, $[a, b] = [0, 1]$ and $\Delta_{x_i} = \Delta_x$ for all $i = 1, 2, \dots, n$, then

$$0 < \frac{1}{n} < \frac{2}{n} < \dots < \frac{n}{n} = 1.$$

When $f(x)$ is integrable on $[0, 1]$, we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left(f\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right) + \dots + f\left(\frac{n}{n}\right) \right) = \int_0^1 f(x) dx.$$

Remark 2

If $f(x)$ is continuous, it can be proved that the limit in the definition is defined. So, f is integrable on $[a, b]$.

3.1.2. Properties

Let $f(x), g(x)$ be integrable on $[a, b]$. Then, we have

$$+ \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

$$+ \int_a^b (f(x) - g(x)) dx = \int_a^b f(x) dx - \int_a^b g(x) dx.$$

$$+ \int_a^b (kf(x)) dx = k \int_a^b f(x) dx \text{ where } k = \text{const.}$$

$$+ \text{If } f(x) \leq g(x) \forall x \in [a, b], \text{ then } \int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

3.1.2. Properties

$$+ \int_a^b (kf(x))dx = k \int_a^b f(x)dx \text{ where } k = \text{const.}$$

$$+ \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \quad \forall c \in (a, b).$$

$$+ \int_a^b f(x)dx = - \int_b^a f(x)dx.$$

$$+ \left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx.$$

$$+ \int_a^b f(x)dx = \int_a^b f(t)dt \quad (\text{it does not depend on the variables name } x).$$

Cauchy's inequality

Let $f(x), g(x)$ be continuous on $[a, b]$. Then,

$$\left(\int_a^b f(x)g(x)dx \right)^2 \leq \int_a^b f^2(x)dx \cdot \int_a^b g^2(x)dx.$$

The midpoint rule (The Mean Value Theorem for Integrals)

If $f(x)$ is continuous on $[a, b]$ and $m \leq f(x) \leq M \quad \forall x \in [a, b]$, then there is a number $\mu \in [m, M]$ such that

$$\mu = \frac{1}{b-a} \int_a^b f(x) dx.$$

3.1.3. Evaluation Theorem (Newton-Leibnitz)

If f is continuous on the closed interval $[a, b]$, then

$$\int_a^b f(x)dx = F(b) - F(a),$$

where F is any antiderivative of f , that is $F'(x) = f(x)$. Note that the family of all antiderivatives of $f(x)$ is

$$\int f(x)dx = F(x) + C \Leftrightarrow F'(x) = f(x).$$

Antiderivative form of indefinite integral

$$\int x^{\alpha} dx = \frac{1}{\alpha + 1} x^{\alpha+1} + C \quad \text{với } \alpha \neq -1, C = \text{const.}$$

$$\int \frac{1}{x} dx = \ln |x| + C.$$

$$\int e^x dx = e^x + C.$$

$$\int a^x dx = \frac{1}{\ln a} a^x + C \quad \text{với } 1 \neq a > 0.$$

$$\int \sin x dx = -\cos x + C.$$

$$\int \cos x dx = \sin x + C.$$

$$\int \frac{1}{\sin^2 x} dx = -\cot x + C.$$

Antiderivative form (continuity)

$$\int \frac{1}{\cos^2 x} dx = \tan x + C.$$

$$\int \operatorname{sh} x dx = \operatorname{ch} x + C.$$

$$\int \operatorname{ch} x dx = \operatorname{sh} x + C.$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C.$$

$$\int \frac{1}{\sqrt{1+x^2}} dx = \ln(x + \sqrt{1+x^2}) + C.$$

$$\int \frac{1}{\sqrt{x^2-1}} dx = \ln|x + \sqrt{x^2-1}| + C.$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C.$$

3.1.4. Computational methods

The Substitution Rule

Theorem

Let the function $f(x)$ be continuous on $[a, b]$ and the function $\varphi(t)$ satisfies follows:

- + $\varphi(\alpha) = a, \varphi(\beta) = b$*
- + $\varphi'(t)$ is continuous on $[\alpha, \beta]$*
- + $\forall t \in [\alpha, \beta] \Rightarrow x = \varphi(t) \in [a, b]$.*

Then,

$$\int_a^b f(x)dx = \int_{\alpha}^{\beta} f(\varphi(t))\varphi'(t)dt.$$

Examples

Calculate the following integrals:

$$I = \int_0^3 \sqrt{9 - x^2} dx.$$

$$J = \int_0^{\ln 3} \frac{e^x dx}{\sqrt{(1 + e^x)^3}}.$$

Examples (continuity)

Let $f(x), g(x)$ be continuous on $\mathcal{R}$.

i) Prove:

$$\int_0^{\frac{\pi}{2}} f(\sin x) dx = \int_0^{\frac{\pi}{2}} f(\cos x) dx.$$

Applying for $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\sin x}}{\sqrt[3]{\sin x} + \sqrt[3]{\cos x}} dx$.

ii) Prove:

$$\int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx.$$

Applying for $I_2 = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

Integration by Parts

Let $u = u(x), v = v(x)$ be continuous on $[a, b]$. From the form

$$d(uv) = u dv + v du,$$

it follows *formula for integration by parts*:

$$\int_a^b u dv = (uv) \Big|_a^b - \int_a^b v du.$$

Example

Calculate

$$I = \int_0^3 \arcsin \sqrt{\frac{x}{1+x}} dx.$$

Solution: Set $u = \arcsin \sqrt{\frac{x}{1+x}}$, $dv = dx$. Using the formula $(\sqrt{u})'_x = \frac{1}{2\sqrt{u}} \cdot u'_x$, $(\arcsin u)'_x = \frac{1}{\sqrt{1-u^2}} \cdot u'_x$, we have $du = u'_x dx$, $v = x + 1$ and

$$u'_x = \frac{1}{\sqrt{1 - \frac{x}{x+1}}} \cdot \frac{1}{2\sqrt{\frac{x}{x+1}}} \left(\frac{x}{x+1} \right)' = \frac{1}{2\sqrt{x}} \cdot \frac{1}{(x+1)}.$$

Then, we get

$$I = (x+1) \arcsin \sqrt{\frac{x}{1+x}} \Big|_0^3 - \int_0^3 \frac{1}{2\sqrt{x}} dx = \dots = \frac{4\pi}{3} - \sqrt{3}.$$

Additional Techniques of Integration

Rational Functions

General, we need to calculate the integral (is a ratio of two polynomials):

$$I = \int_a^b \frac{P_n(x)}{Q_m(x)} dx,$$

where $P_n(x)$, $Q_m(x)$ are polynomials n -degree and m -degree respectively.

Additional Techniques of Integration

Basis Fractions

$$I_1 = \int_{\alpha}^{\beta} \frac{dx}{ax+b} \quad (a \neq 0).$$

$$I_2 = \int_{\alpha}^{\beta} \frac{dx}{(ax+b)^{\gamma}} \quad (a \neq 0, \gamma \neq 1).$$

$$I_3 = \int_{\alpha}^{\beta} \frac{dx}{x^2+a^2}.$$

$$I_4 = \int_{\alpha}^{\beta} \frac{xdx}{x^2+a^2}.$$

$$I_5 = \int_{\alpha}^{\beta} \frac{dx}{(x^2+a^2)^{\gamma}} \quad (\gamma \geq 2, \gamma \in \mathbb{N}).$$

$$I_6 = \int_{\alpha}^{\beta} \frac{xdx}{(x^2+a^2)^{\gamma}} \quad (\gamma \geq 2, \gamma \in \mathbb{N}).$$

Example

Use the Forms of Integrals to find

$$I = \int_1^{\sqrt{3}} \frac{3x + 1}{x^2(1 + x^2)^2} dx.$$

Trigonometric Integrals

Consider

$$I = \int_{\alpha}^{\beta} R(\sin x, \cos x) dx.$$

General form

Solution: Set $t = \tan \frac{x}{2}$.

Example

Calculate

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{2 \sin x - \cos x + 5}.$$

Trigonometric Integrals

Consider

$$I = \int_{\alpha}^{\beta} R(\sin x, \cos x) dx.$$

Form 1: $R(-\sin x, -\cos x) = R(\sin x, \cos x)$ for all x .

Solution: Set $t = \tan x$.

Example

Calculate

$$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{dx}{\sin^3 x \cos^5 x}.$$

Trigonometric Integrals

Consider

$$I = \int_{\alpha}^{\beta} R(\sin x, \cos x) dx.$$

Form 2: $R(-\sin x, \cos x) = -R(\sin x, \cos x)$ for all x .

Solution: Set $t = \cos x$.

Example

Calculate

$$I = \int_0^{\frac{\pi}{4}} \frac{\sin x dx}{\cos x \sqrt{1 + \sin^2 x}}.$$

Trigonometric Integrals

Consider

$$I = \int_{\alpha}^{\beta} R(\sin x, \cos x) dx.$$

Form 3: $R(\sin x, -\cos x) = -R(\sin x, \cos x)$ for all x .

Solution: Set $t = \sin x$.

Example

Calculate

$$I = \int_0^{\frac{\pi}{2}} \frac{(\sin 2x + \cos x) dx}{\sqrt{1 + 3 \sin x}}.$$

Applications of Integration

First: Areas between Curves

$$(H_1) \begin{cases} y = f(x) \\ y = g(x) \\ x = a \quad (a < b) \\ x = b \end{cases} \Rightarrow S_{H_1} = \int_a^b |f(x) - g(x)| dx.$$

Example

Consider the region that lies between two curves

$$(H) \begin{cases} x + y = 0 \\ y = 2x - x^2. \end{cases}$$

Define the area (H) .

Second: The volume of a solid

$$(H) \begin{cases} y = f(x) \geq 0 & \forall x \in [a, b] \\ x = a, \\ x = b. \end{cases} \Rightarrow V_H = \pi \int_a^b f^2(x) dx.$$

Example

Let $a > 0, b > 0$. Consider the solid defined by

$$(H) : \begin{cases} \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \\ (\alpha) : z = \frac{c}{a}x, \\ (\beta) : z = 0. \end{cases}$$

Calculate the volume of (H) .

Third: Arc length

If a smooth curve with parametric equations

$$\widetilde{AB} : \begin{cases} x = x(t) \\ y = y(t) \end{cases} \quad t_1 \leq t \leq t_2. \quad (1)$$

Then, its length is

$$|\widetilde{AB}| = \int_{t_1}^{t_2} \sqrt{x_t'^2 + y_t'^2} dt.$$

Example

Find the length of Astroid curve:

$$\widetilde{AB} : \begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases} \quad a > 0.$$

Third: Arc length (continuity)

If a smooth curve with equation on (Oxy) :

$$\widetilde{AB} : y = f(x) \quad x \in [a, b].$$

Then, its length is

$$|\widetilde{AB}| = \int_a^b \sqrt{1 + f'(x)^2} dx.$$

Example

Find the length of the curve:

$$\widetilde{AB} : y^2 = 2x, 0 \leq x \leq \frac{1}{2}.$$

Third: Arc length (continuity)

If a smooth curve with equation on cylindrical coordinates

$$\widetilde{AB} : r = r(\varphi) \quad \varphi \in [\alpha, \beta].$$

Then, its length is

$$|\widetilde{AB}| = \int_{\alpha}^{\beta} \sqrt{r^2(\varphi) + r'^2(\varphi)} d\varphi.$$

Example

Find the length of the curve:

$$\widetilde{AB} : r = a(1 + \cos \varphi), a > 0.$$

Fourth: Areas of around face (H)

The face (H) is defined by the curve AB surrounding the axis Ox .

Form 1: The curve AB is given as

$$\widetilde{AB} : y = f(x) \geq 0 \quad a \leq x \leq b.$$

Then, the area of (H) is

$$S_H = 2\pi \int_a^b f(x) \sqrt{1 + f'^2(x)} dx.$$

Fourth: Areas of around face (H) (continuity)

The face (H) is defined by the curve AB surrounding the axis Ox .

Form 2: The curve AB is given as

$$r = r(\varphi) \quad \text{with} \quad \forall \varphi \in [\alpha, \beta].$$

Then, the area of (H) is

$$S_H = 2\pi \int_{\alpha}^{\beta} r(\varphi) \sqrt{r^2(\varphi) + r'^2(\varphi)} d\varphi.$$

Fourth: Areas of around face (H) (continuity)

The face (H) is defined by the curve AB surrounding the axis Ox .

Form 3: The curve AB is given as

$$\widetilde{AB} : \begin{cases} x = x(t) \\ y = y(t) \geq 0 \end{cases} \quad t_1 \leq t \leq t_2.$$

Then, the area of (H) is

$$S_H = 2\pi \int_{t_1}^{t_2} y(t) \sqrt{x'^2(t) + y'^2(t)} dt.$$

Fourth: Areas of around face (H) (continuity)

Example

The face (H) is defined by the curve AB surrounding the axis Ox , where AB is Cycloide curve as follows

$$\widetilde{AB} : \begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \quad 0 \leq t \leq 2\pi, \quad a > 0.$$

Calculate the area of (H).

Integral Calculus

1 3.1. The Definite Integral

2 3.2. Improper Integrals

Type 1: Infinite Integrals

Definition

- If exists $\int_a^y f(x)dx$ for every number $y > a$, then

$$\int_a^{+\infty} f(x)dx = \lim_{y \rightarrow +\infty} \int_a^y f(x)dx.$$

provided this limit exists (as a finite number).

- If exists $\int_y^b f(x)dx$ for every number $y < b$, then

$$\int_{-\infty}^b f(x)dx = \lim_{y \rightarrow -\infty} \int_y^b f(x)dx.$$

Type 1: Infinite Integrals (continuity)

Definition

The improper integrals $\int_a^{+\infty} f(x)dx$ and $\int_{-\infty}^b f(x)dx$ are called convergent if the corresponding limit exists and divergent if the limit does not exist.

- If both $\int_a^{+\infty} f(x)dx$ and $\int_{-\infty}^a f(x)dx$ are convergent, then we define

$$\int_{-\infty}^{+\infty} f(x)dx = \int_{-\infty}^a f(x)dx + \int_a^{+\infty} f(x)dx.$$

Type 1: Infinite Integrals (continuity)

Example

Let $a > 0$. Evaluate the following infinite integrals.

$$(i) \quad I_1 = \int_0^{+\infty} \frac{dx}{x^2+a^2}.$$

$$(ii) \quad I_2 = \int_{-\infty}^0 \frac{x dx}{x^2+a^2}.$$

$$(iii) \quad I_3 = \int_{-\infty}^{+\infty} \frac{dx}{\sqrt{x^2+a^2}}.$$

$$(iv) \quad I_4 = \int_1^{+\infty} \frac{dx}{x^\alpha} \quad \text{where constant } \alpha \in \mathbb{R}.$$

Convergent Condition of Infinite Integrals

Comparison Theorem

Suppose that f and g are continuous functions such that $f(x) \geq g(x) \geq 0$ for all $x \geq a$. Then, we have

- (a) If $\int_a^{+\infty} f(x)dx$ is convergent, then $\int_a^{+\infty} g(x)dx$ is convergent.
- (b) If $\int_a^{+\infty} g(x)dx$ is divergent, then $\int_a^{+\infty} f(x)dx$ is divergent.

Example

By using the Comparison Theorem, determine whether

$$\int_1^{+\infty} \frac{x dx}{2x^5 + x^2 - 1}.$$

it converges or diverges.

Convergent Condition of Infinite Integrals (continuity)

Limit Theorem

Suppose that $f(x) \geq 0, g(x) \geq 0$ for all $x \geq a$ and

$$\lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = M.$$

(a) In the case $M \in (0, +\infty)$. Then, $\int_a^{+\infty} f(x)dx$ is convergent if and

only if $\int_a^{+\infty} g(x)dx$ is convergent.

(b) In the case $M = 0$. If $\int_a^{+\infty} g(x)dx$ is convergent then $\int_a^{+\infty} f(x)dx$ is also convergent.

(c) In the case $M = +\infty$. If $\int_a^{+\infty} g(x)dx$ is divergent then $\int_a^{+\infty} f(x)dx$ is also divergent.

Convergent Condition of Infinite Integrals (continuity)

Example

Determine whether each integral is convergent or divergent:

$$I_1 = \int_1^{+\infty} \frac{(2x^2 + 1)dx}{x^3 \sqrt[3]{x^4 + 2x + 1}}, \quad I_2 = \int_3^{+\infty} \frac{\ln(x - 2)dx}{3x^3 + x^2 + 1}.$$

Absolute Convergence of Infinite Integrals

Definition

Infinite Integral $\int_a^{+\infty} f(x)dx$ is called *absolutely convergent*, if

$\int_a^{+\infty} |f(x)|dx$ is convergent.

+ The integral $\int_a^{+\infty} f(x)dx$ is called *semi-convergent*, if $\int_a^{+\infty} f(x)dx$ is convergent and $\int_a^{+\infty} |f(x)|dx$ is divergent.

Absolute Convergence of Infinite Integrals (continuity)

Theorem

If $\int_a^{+\infty} f(x)dx$ is absolutely convergent, then it is convergent.

Example

Show that the integral

$$I = \int_1^{+\infty} \frac{\sin x}{x} dx$$

is semi-convergent and not absolutely convergent.

Type 2: Discontinuous Integrals

Definition

- If $f(x)$ is continuous on $[a, b)$ and $\lim_{x \rightarrow b^-} f(x) = \infty$, then

$$\int_a^b f(x)dx = \lim_{y \rightarrow b^-} \int_a^y f(x)dx.$$

if this limit exists (as a finite number).

- If $f(x)$ is continuous on $(a, b]$ and $\lim_{x \rightarrow a^+} f(x) = \infty$, then

$$\int_a^b f(x)dx = \lim_{y \rightarrow a^+} \int_y^b f(x)dx.$$

if this limit exists (as a finite number).

Type 2: Discontinuous Integrals (continuity)

Example

Let $a > 0$. Evaluate the following infinite integrals

$$I = \int_a^b \frac{dx}{(x-a)^\alpha} \quad (a < b, \alpha \in \mathbb{R}).$$

Convergent Condition of Discontinuous Integrals

Comparison Theorem

Suppose that f and g are continuous functions such that $f(x) \geq g(x) \geq 0$ for all $x \in [a, b)$, $\lim_{x \rightarrow b^-} f(x) = \infty$ and $\lim_{x \rightarrow b^-} g(x) = \infty$.

Then, we have

(a) If $\int_a^b f(x)dx$ is convergent, then $\int_a^b g(x)dx$ is convergent.

(b) If $\int_a^b g(x)dx$ is divergent, then $\int_a^b f(x)dx$ is divergent.

Example

By using the Comparison Theorem, determine whether

$$I = \int_0^1 \frac{|\ln x|}{x^\alpha} dx \quad (\alpha \in \mathbb{R}).$$

Convergent Condition of Discontinuous Integrals (continuity)

Limit Theorem

Suppose that $f(x) \geq 0, g(x) \geq 0$ for all $x \in [a, b)$, $\lim_{x \rightarrow b^-} f(x) = \infty$,
 $\lim_{x \rightarrow b^-} g(x) = \infty$ and $\lim_{x \rightarrow b^-} \frac{f(x)}{g(x)} = M$ (it follows $M \in [0, +\infty]$).

(a) In the case $M \in (0, +\infty)$: $\int_a^b f(x)dx$ is convergent if and only if

$\int_a^b g(x)dx$ is convergent.

(b) In the case $M = 0$. If $\int_a^b g(x)dx$ is convergent then $\int_a^b f(x)dx$ is also convergent.

(c) In the case $M = +\infty$. If $\int_a^b g(x)dx$ is divergent then $\int_a^{+\infty} f(x)dx$ is also divergent.

Convergent Condition of Discontinuous Integrals (continuity)

Example

Determine whether each integral is convergent or divergent:

$$I_1 = \int_0^1 \frac{dx}{\sqrt{(1-x^2)(1-k^2x^2)}} \quad \text{where } |k| < 1.$$

$$I_2 = \int_0^1 \frac{dx}{\sqrt[3]{x(e^x - e^{-x})}}.$$

$$I_3 = \int_0^{\frac{\pi}{2}} \frac{\ln(\sin x)}{\sqrt{x}} dx.$$